Media Engineering and Technology Faculty
German University in Cairo





Sentiment-Enhanced Deep Stock Market Forecasting Using Bilingual Financial News: An Egyptian Market Case Study

Bachelor Thesis

Author: Hussien Haitham Hussien Abdelmotaleb Hussien
Supervisors: Dr. Mervat AbuElkheir
Submission Date: 8 June 2026

Media Engineering and Technology Faculty
German University in Cairo





Sentiment-Enhanced Deep Stock Market Forecasting Using Bilingual Financial News: An Egyptian Market Case Study

Bachelor Thesis

Author: Hussien Haitham Hussien Abdelmotaleb Hussien
Supervisors: Dr. Mervat AbuElkheir
Submission Date: 8 June 2026

This is to certify that:

(i) the thesis comprises only my original work toward the Bachelor Degree

(ii) due acknowlegement has been made in the text to all other material used

Disclosure of Generative AI Assistance.
Generative AI tools (Anthropic Claude and Claude Code) were used for: brainstorming experiment structure, drafting and refining LaTeX prose, generating boilerplate code for evaluation scripts, and auditing the manuscript against a style checklist. All scientific decisions, experimental design, data handling, result interpretation, and final wording were performed and verified by the author. No AI tool was used to fabricate data, results, or citations.

Hussien Haitham Hussien Abdelmotaleb Hussien
8 June, 2026

Acknowledgments

This thesis would not exist without the people around me, and I want to take a moment to actually say that.

To my family — my mother, my father, and my sister — thank you for everything. The support, the patience, the belief even when I had none. This is as much yours as it is mine.

To my friends, thank you for being there throughout. You know who you are and what it meant.

To the CSIS group: you are my brothers. There is no version of this where I get here without you. Every late night, every shared frustration, every moment of pushing each other forward — it all added up. I am genuinely grateful.

To my supervising team at Robin — Mostafa El Far, Logine El Kelany, and Abdelrahman Hassan — thank you for the guidance, the feedback, and for taking this project seriously. Working with you made the work better.

A special thank you to Omar Kamal, who did the interviews at the very beginning and believed in what this could become before it became anything. That belief mattered more than I can properly express.

Thank you to Basma Rady, Managing Director of Robin, for creating an environment where a project like this was even possible. And thank you to the entire Robin team for the collaboration and the warmth throughout.

Abstract

This thesis will investigate whether bilingual financial-news sentiment can support short-horizon stock direction forecasting on the Egyptian Exchange (EGX). The empirical scope will cover three EGX equities: Commercial International Bank (CIB), Talaat Moustafa Group (TMG), and Orascom Investment Holding (OIH). The article corpus will be drawn from Mubasher company pages and will include 1,921 bilingual articles after cleaning, with 1,335 Arabic items and 586 English items spanning a multi-year window. The motivation will be an industrial brief from Robin, the data-science arm of Beltone Holding, which will frame the question of whether sentiment-derived signals will add forecasting value beyond historical price data while remaining interpretable and reproducible.

The pipeline will collect bilingual articles, route English text to FinBERT and Arabic text to an AraBERT-based sentiment route, weight each article by ticker-match confidence and model confidence, aggregate article-level scores into one-day-lagged daily features, and evaluate all-news, event-only, and event-weighted representations under chronological walk-forward validation. The final sentiment-only benchmark will use logistic regression for direction prediction and ridge regression for future-return prediction at horizons of 1, 3, and 5 trading days. Leakage, coverage, and overfitting audits will form part of the experimental design rather than serve as optional post-processing.

The reporting plan will prefer conservative, validity-passing results over the highest raw accuracy. The accepted final candidate will be reported per ticker, horizon, and feature view together with balanced accuracy, F1, return sign accuracy, and the train-validation-test gap that justifies acceptance. Four auxiliary experiments will accompany the main benchmark: a historical-model integration test with sentiment as an added feature, a Spearman correlation study with five-day block-permutation p-values, an Arabic financial-domain Masked Language Modeling (MLM) adaptation of AraBERT measured by held-out MLM perplexity (this measures domain language modeling fit, not supervised sentiment performance, which will remain future work pending Arabic financial sentiment labels), and a social-media access feasibility audit covering Reddit and X/Twitter.

The methodological contribution will be a reproducible, language-aware, and leakage-safe protocol for testing bilingual financial-news sentiment in a sparse emerging-market setting. All scripts, data manifests, evaluation outputs, and audit reports will be released through a public repository so that every reported number can be regenerated from the same configuration.

Contents

Acknowledgments **V**

1 Introduction **1**
1.1 Problem Statement . . . 2
1.2 Motivation . . . 2
1.3 Research Questions . . . 2
1.4 Objectives . . . 2
1.5 Thesis Outline . . . 3

2 Background **5**
2.1 Financial Forecasting as a Problem of Limited Predictability . . . 5
2.2 Forecast Targets and Input Modalities . . . 5
2.3 Class Imbalance and Evaluation Sensitivity . . . 6
2.4 Textual Information and Sentiment in Finance . . . 6
2.5 From Lexicons to Deep Financial NLP . . . 7
2.6 Arabic and Bilingual Financial Text Processing . . . 8
2.7 Chronological Evaluation and Leakage . . . 8
2.8 Chapter Summary . . . 9

3 Related Work **11**
3.1 From Price-Only to Text-Aware Forecasting . . . 11
3.2 Social Media Sentiment . . . 11
3.3 Financial News Sentiment . . . 12
3.4 Why Sentiment Does Not Always Improve Forecasting . . . 13
3.5 Domain-Specific and Arabic Language Models . . . 15
3.6 Evaluation Metrics and Research Gap . . . 16
3.7 Chapter Summary . . . 16

4 Methodology **17**
4.1 Research Design . . . 17
4.2 Study Universe and Data Sources . . . 18
4.3 News Collection and Article Preparation . . . 18
4.4 Bilingual Sentiment Extraction . . . 19
4.5 Confidence Filtering and Daily Sentiment Aggregation . . . 20
4.6 Event-Aware Feature Variants . . . 21
4.7 Target Construction and Lagged Alignment . . . 21
4.8 Forecasting Framework . . . 22
4.9 Evaluation Metrics . . . 22
4.10 Leakage Control and Validity Checks . . . 23

4.11 Auxiliary Experiment Protocols . . . 23
4.12 Chapter Summary . . . 24

5 Results & Implementation **25**
5.1 Experimental Setup . . . 25
5.2 Final Experimental Dataset . . . 27
5.3 Main Multi-Horizon Results . . . 27
5.4 Event Filtering and Event Weighting . . . 28
5.5 Validity Checks and Accepted Final Model . . . 29
5.6 Auxiliary Experiments and Research Journey . . . 29
5.7 Interpretation . . . 31
5.8 Chapter Summary . . . 32

6 Conclusion & Future Work **33**
6.1 Conclusion . . . 33
6.2 Future Work . . . 34

Appendix **36**

A Lists **39**
List of Abbreviations . . . 39
List of Figures . . . 41

References **45**

Chapter 1

Introduction

Stock markets are not only trading venues. They are mechanisms for capital allocation, corporate financing, and risk transfer. When public information is absorbed into prices, the speed and quality of that absorption affect investors, listed firms, and the broader economy [1, 2, 3]. For research desks and brokerage institutions, the practical question is therefore not merely whether prices move. The question is whether new information can be interpreted quickly enough to support better decisions before the reaction is fully reflected in the market. This economic relevance is one reason why stock forecasting remains a persistent research problem.

The technical difficulty begins with an information asymmetry. Numerical market series record the market reaction after it occurs. News, disclosures, and commentary describe the events that may have triggered that reaction. A price chart reveals that a move happened. It does not explain whether the move followed earnings news, financing plans, governance changes, or contract announcements. This gap is why researchers have turned to text as an additional signal on top of price history [4, 5, 3].

Existing research therefore moves from price-only forecasting toward multimodal forecasting that combines market variables with textual sentiment, event representations, or contextual language-model features. The evidence is encouraging but not uniform. Some studies report measurable gains from financial news or social-media sentiment. Others show that those gains depend strongly on the text source, the forecast horizon, the market under study, and the evaluation protocol [6, 7, 8, 9]. The relevant scientific question is conditional: under what data and validation conditions does sentiment add predictive value?

This question is especially important on the Egyptian Exchange (EGX), where firm-related financial coverage appears in both Arabic and English. A forecasting pipeline that ignores one of the two languages discards part of the information environment. A pipeline that combines both languages without careful routing and relevance control can instead amplify noise. This thesis grew out of a bachelor project with Robin, the data-science arm of Beltone Holding, where this issue is practical rather than abstract. The working need is a test of whether bilingual financial-news sentiment supports EGX-oriented forecasting analysis under a setting that remains interpretable, conservative, and reproducible. Prior work on finance-specific English sentiment modelling, Arabic-specialised language models, and conservative sentiment-evaluation design supports an emphasis on language-aware modelling and strict temporal alignment [10, 11, 12, 9].

1.1 Problem Statement

This thesis investigates whether sentiment extracted from bilingual financial news improves short-horizon stock direction forecasting in order to determine the predictive value of sentiment-derived signals for three EGX equities: Commercial International Bank (CIB), Talaat Moustafa Group (TMG), and Orascom Investment Holding (OIH).

1.2 Motivation

The industrial context of the study motivates the chosen design. In the Robin and Beltone workflow that motivates the thesis, analysts face a mixed Arabic-English news stream, uneven company coverage, and many articles that are attached to a company page without being equally relevant to valuation. A forecasting signal in that setting must do more than assign positive or negative labels. It must route each language to an appropriate sentiment model. It must distinguish direct company coverage from weaker mentions. It must aggregate articles in a way that preserves timing and relevance. This strategy is consistent with prior work showing that source choice, finance-specific English sentiment modelling, and Arabic-specialised encoders materially affect the quality of textual sentiment signals [5, 10, 11].

A second motivation is methodological. A strong price-based forecasting system can hide whether sentiment is informative or whether it is merely an additive feature inside an already expressive model. The thesis therefore adopts a sentiment-only benchmark before any later integration with richer market features. This choice makes the interpretation cleaner. The choice also aligns with the cautionary literature on sentiment forecasting, where apparent gains often disappear once feature leakage, horizon choice, or loose validation procedures are controlled more strictly [9, 8, 3].

1.3 Research Questions

RQ1. To what extent does bilingual financial-news sentiment contain predictive information for short-horizon EGX stock direction forecasting?

RQ2. How do language-specific routing, article relevance, and event-aware aggregation affect the quality of the extracted sentiment signal?

RQ3. Do any observed predictive gains remain valid under strictly chronological and leakage-safe evaluation, and what do auxiliary experiments with historical-model integration, sentiment-return correlation, Arabic financial-domain adaptation, and social-media feasibility reveal about the future development path of EGX sentiment forecasting?

1.4 Objectives

Objective 1. Design a bilingual financial-news forecasting formulation for the three target EGX equities.

Objective 2. Develop a sentiment extraction framework that distinguishes Arabic and English article processing through language routing.

Objective 3. Construct sentiment features that account for article relevance, language composition, and event structure.

Objective 4. Evaluate the resulting forecasting signal under chronological validation and test splits with explicit leakage controls.

Objective 5. Implement and document four auxiliary protocols (historical-model integration, sentiment-return association, Arabic financial-domain adaptation, social-media feasibility) so that the research journey is auditable.

Objective 6. Analyse whether the observed sentiment signal is strong enough to justify further sentiment-enhanced forecasting work in the Egyptian market and recommend next steps for richer multimodal forecasting.

1.5 Thesis Outline

The remainder of this thesis is organised as follows. Chapter 2 establishes the background concepts required for the study: forecasting targets, class imbalance in directional prediction, financial text as an exogenous information source, transformer-based sentiment extraction, and leakage-safe evaluation. Chapter 3 reviews state-of-the-art work on sentiment-based stock forecasting and positions the present thesis within that literature. Chapter 4 presents the solution approach adopted in the study. Chapter 5 reports the implementation choices, experimental setup, and empirical findings. Chapter 6 summarises the main conclusions and outlines future research directions.

Chapter 2

Background

This chapter covers the building blocks the rest of the thesis depends on. It starts with why stock forecasting is hard, then moves through forecast target definitions, class imbalance in directional prediction, textual signals in finance, the shift from word lists to deep NLP models, Arabic and bilingual text processing, and evaluation discipline. These concepts are what Chapter 4 is built on.

2.1 Financial Forecasting as a Problem of Limited Predictability

Stock forecasting sits at the intersection of economics, statistics, and machine learning. Its difficulty begins with the nature of the market itself. Financial prices respond to heterogeneous information, strategic trading, liquidity effects, and macroeconomic shocks. The efficient-market literature formalised this challenge by arguing that public information is rapidly incorporated into prices. Later empirical work showed that the random-walk view is too strict to hold universally, but it also confirmed that predictable structure in market data is weak, unstable, and difficult to exploit consistently [1, 2].

This tension explains why forecasting remains an active research area. If markets were perfectly predictable, the problem would already be solved. If they were perfectly unpredictable, forecasting models would be pointless. The research problem lies between these extremes. Small predictive patterns may exist, but they are noisy, regime-dependent, and sensitive to how the prediction target is defined. Surveys of the field show that forecasting studies differ substantially in horizon, label construction, feature design, and evaluation protocol. Those choices often matter as much as the choice of learning algorithm itself [13, 3].

2.2 Forecast Targets and Input Modalities

Financial forecasting studies use several related targets. Some studies predict the next price level. Others predict return magnitude, return sign, or discrete upward versus downward direction. Directional forecasting aligns naturally with decision-making, but

it discards part of the information contained in the return magnitude. Regression targets preserve more information yet are often harder to estimate accurately. Many studies therefore evaluate both direction and return as complementary views of the same forecasting task [3, 14, 15].

The corresponding input spaces also vary. Traditional models rely on historical prices, returns, volume, and hand-crafted technical indicators. These variables summarise past market behaviour, but they do not describe the semantic content of new information. A news article about profits, financing, or governance may precede a price move even when the numerical history alone is ambiguous. This observation motivates the use of exogenous text signals in addition to market data. The broader forecasting literature increasingly treats stock prediction as a multimodal problem rather than a purely numerical one [13, 3, 14].

2.3 Class Imbalance and Evaluation Sensitivity

Directional forecasting introduces a statistical issue that is easy to underestimate: class imbalance. Upward and downward moves are rarely distributed evenly across a financial sample. Persistent market drift, low-volatility periods, and sparse firm-level event coverage can all push the labels toward one class. In that setting, raw accuracy overstates model quality because a classifier can appear successful while still failing on the minority class. This thesis therefore reports balanced accuracy, F1, and confusion-matrix statistics alongside headline accuracy, in line with the evaluation caution emphasised in recent forecasting studies [3, 9, 16].

The problem becomes sharper in sentiment-based firm-level studies. News coverage is uneven across firms, and longer forecast horizons can create clusters of same-direction labels. A model then learns the dominant class more easily than it learns a robust relation between sentiment and future movement. Table 2.1 summarises why imbalance matters and which safeguards are needed before a result can be interpreted as economically meaningful.

2.4 Textual Information and Sentiment in Finance

Financial researchers treat news and corporate disclosures as meaningful signals, not as background noise. Early work examined whether media tone, message-board language, or corporate disclosures explain subsequent market behaviour. Negative media language correlated with market activity and price pressure [4]. Message-board sentiment also contained information about volume and volatility, although the signal quality depended on aggregation and filtering [6]. Finance-specific dictionaries later showed that general-purpose sentiment lexicons can misclassify domain vocabulary, which is why terms such as “liability” or “capital” require financial context to be interpreted correctly [17, 5].

These findings motivate a distinction between sentiment source and sentiment quality. Not every text stream has the same predictive value. Financial news is more structured than social-media text and is more likely to contain firm-linked events. Social media is timelier and larger in volume, but it is also noisier and harder to map to a specific

Table 2.1: Why class imbalance is a methodological concern in directional financial forecasting.

Imbalance source	Financial forecasting manifestation	Risk if ignored	Appropriate safeguard
Market drift	One direction dominates over a sample period, especially in short binary up/down labels	Raw accuracy overstates true discrimination	Balanced accuracy and confusion-matrix reporting
Sparse firm coverage	Company panels with few relevant articles generate unstable minority-class observations	High F1 can be driven by majority-class prediction	Per-ticker evaluation and coverage analysis
Longer target horizons	Multi-day returns can cluster on one side during sustained moves	Thresholds become biased toward the dominant class	Validation-only threshold tuning and horizon-specific reporting
Event concentration	Market-moving news arrives in bursts rather than uniformly across dates	A model can memorise event periods instead of learning a stable relation	Chronological splits with embargo and walk-forward validation

asset. Survey work in financial sentiment analysis emphasises that source choice, content analysis method, and aggregation strategy jointly determine whether textual sentiment becomes a useful explanatory variable [5, 18, 7].

2.5 From Lexicons to Deep Financial NLP

Text-based market prediction evolved from sparse lexical features toward distributed and contextual representations. Early financial news systems used bag-of-words, noun phrases, named entities, and support-vector models. The AZFinText system used breaking financial news and intra-day stock data to predict short-term price movement, showing that structured textual features support directional prediction [19]. A later context-capturing study reported improvements over simpler textual baselines for stock-price prediction from financial news [20]. These systems demonstrated that the representation of text strongly influences the usefulness of the downstream forecasting model.

Deep learning extended this line of work by learning latent semantic features directly from text. Event-driven approaches represented news events in dense form and modelled how short-term and long-term event effects influence price movement [21]. Other models fused news with recurrent architectures, attention mechanisms, or hybrid text-price representations [22, 23, 24]. These studies moved the field from manually designed textual cues toward learned semantic encodings. The same studies made clear that the meaning of a financial article cannot be reduced to isolated polarity words alone.

The broader NLP shift toward transformer-based models is also central to this thesis. The transformer architecture replaced recurrent sequence processing with self-attention, which enables stronger contextual representations at scale [25]. The Bidirectional Encoder Representations from Transformers (BERT) model then showed how pre-trained bidirectional language models can be adapted to downstream tasks with minimal architectural change [26]. Domain-specialised variants followed, including FinBERT for English financial text [10]. This progression matters because sentiment extraction quality increasingly depends on pre-training domain and language rather than on task-specific feature engineering alone.

2.6 Arabic and Bilingual Financial Text Processing

Arabic financial NLP introduces additional complexity. Arabic is morphologically rich, orthographically variable, and distributed across Modern Standard Arabic, dialectal Arabic, and specialised domain language. These properties make direct transfer from English sentiment models unreliable. Arabic-specific pre-training is therefore important for downstream classification quality. AraBERT demonstrated strong gains over multilingual baselines on Arabic NLP tasks [11]. ARBERT and MARBERT showed that specialised Arabic pre-training outperforms larger multilingual models on a broad benchmark of Arabic understanding tasks [12].

Bilingual settings introduce a further modelling decision: whether to force all text through a single multilingual model or to route each language through a more suitable encoder. Recent Arabic model studies indicate that language variant and task proximity are more important than raw pre-training size [27, 12, 11]. For a bilingual financial-news pipeline, these findings justify treating language routing as a methodological design choice rather than a preprocessing detail.

2.7 Chronological Evaluation and Leakage

The final background concept is evaluation discipline. Time-series forecasting is unusually vulnerable to leakage because future information can enter the feature space in subtle ways. Leakage may come from same-day article alignment, overlapping target windows, hyperparameter tuning on test periods, or refitting logic that uses observations whose targets are not yet known at prediction time. Reported performance can therefore be overstated even when the implementation looks reasonable on the surface [9, 8].

This concern is especially relevant for text-based forecasting. News has timestamps, trading occurs on exchange calendars, and forecast horizons span multiple future observations. A method that ignores those interactions can produce optimistic but invalid results. Chronological train-validation-test splits, embargo regions between adjacent blocks, and walk-forward evaluation are not optional refinements. They are part of the problem definition itself [9, 3]. This thesis adopts that view and treats leakage control as a core experimental requirement rather than as a final diagnostic step.

2.8 Chapter Summary

This chapter established the conceptual foundations for the thesis. Stock forecasting is difficult because market predictability is limited and unstable. Textual information attracts attention because it describes events that numerical histories do not represent directly. The usefulness of that information depends on representation quality, label balance, language suitability, and evaluation discipline. These concepts feed directly into the literature review in the next chapter, which looks at how prior work put them into practice and where this thesis adds something new.

Chapter 3

Related Work

This chapter surveys prior work on sentiment-based stock forecasting. It is organised around four threads: the shift from price-only to text-aware prediction, social-media sentiment, financial-news sentiment, and domain-specific or Arabic language models. Each section ties results from the original work back to the bilingual EGX context. The chapter closes by discussing where sentiment forecasting tends to break down and where this thesis fits in.

3.1 From Price-Only to Text-Aware Forecasting

Stock-prediction research did not begin with sentiment. Early systems relied on prices, returns, volume, and technical indicators because those variables are directly observable and easy to align temporally. The main weakness of that tradition is explanatory rather than computational. Numerical series encode market state, but they do not encode the semantic content of the events that moved the market. Once researchers began treating news and social media as external information channels, forecasting turned into a multimodal problem rather than a purely numerical one [14, 15, 3].

The shift from price-only to text-aware prediction followed a recognisable sequence. The AZFinText system used breaking financial news to support short-term directional prediction and showed that textual features add value when the entity linkage is explicit and the horizon is short [19]. A later context-capturing study improved on bag-of-words pipelines for financial news classification [20]. Subsequent work replaced manual text features with event embeddings, hierarchical attention, and latent-variable sequence models [21, 22, 24]. Structured event representations modelled how short-term and long-term event effects influenced price movement [21]. A hierarchical attention model captured news sequences [22]. A deep generative framework fused tweets and price history and reported new state-of-the-art results on a newly collected dataset [24]. The common lesson is that text representation, entity resolution, and temporal alignment are as important as the downstream predictor.

3.2 Social Media Sentiment

Social media attracted sustained attention because it is timely, large in volume, and rich in investor reaction. A landmark mood-based study analysed approximately 9.8

million tweets from February to December 2008 and linked extracted mood dimensions to subsequent movements in the Dow Jones Industrial Average (DJIA) [7]. The best reported directional result reached 87.6% accuracy after including mood information. The setting is index-level rather than firm-level, so aggregate mood can correlate with broad market movement without solving the harder task of mapping noisy user text to specific equities.

A later message-board study moved closer to firm-level prediction by combining company-related posts with topic-aware sentiment features [18]. The experiment covered 18 stocks over one year, with 171 transaction dates for training and 78 for testing. The topic-sentiment method improved mean accuracy by 2.07 percentage points across the 18 stocks and by 9.83 percentage points on the subset of difficult stocks relative to a price-only baseline. This result quantifies the value of better ticker-specific filtering. The setup remains short-horizon, single-language, and dependent on shallow topic extraction rather than contextual language models.

A multi-signal comparison study evaluated survey sentiment, news headlines, Twitter data, and search-engine signals in the same predictive setting [28]. The main reported finding is that different sources behave differently in time, with search and social signals more informative for short-run market behaviour than traditional survey measures. Sentiment source is therefore not an interchangeable design choice. Speed, topic granularity, and user noise all affect what a sentiment variable actually measures.

Taken together, these social-media studies show that sentiment only carries useful information after heavy filtering. Short text length, sarcasm, reposted content, unofficial company references, and bot-driven amplification all weaken the link between message sentiment and tradable firm-specific information [7, 18, 29]. These constraints become sharper for emerging markets with bilingual discussion, because entity linking and language variation must be solved before modelling begins.

Arabic social-media forecasting work is especially relevant to the future direction of this thesis. A correlation and Granger-causality study examined Arabic tweets and Arab stock-market indices, including EGX30 [30]. The reported associations support the idea that Arabic investor discussion may carry market-relevant information for Egyptian equities. The same study reinforces the need for careful source access and entity filtering. Social-media relevance depends on whether the collected posts can be legally obtained, temporally aligned, and linked to the correct market or company.

3.3 Financial News Sentiment

Financial news is a cleaner signal source than social media because the language is more structured and the company linkage is more explicit. A large-scale S&P 500 panel study assembled five years of daily prices for S&P 500 firms together with more than 265,000 related financial news articles [23]. Deep models built from price data, sentiment information, and hybrid text-price inputs were then compared. The scale of the dataset is a strength because many earlier sentiment studies were limited by sparse article counts. External validity is limited, because a large U.S. large-cap news panel is an easier information environment than a thinner bilingual EGX news stream.

A controlled architecture study compared LSTM and GRU models under identical conditions with and without financial news sentiment [8]. Performance was assessed with

Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the correlation coefficient R. The sentiment-augmented variants outperformed stock-only variants, which isolates the effect of adding text. The study focuses on regression quality for developed-market data, not on leakage-sensitive directional classification for sparse firm-level coverage.

A multi-representation fusion study combined sentiment analysis with candlestick-chart representations in a two-branch Convolutional Neural Network (CNN) [31]. Experiments covered five high-demand U.S. stocks, namely Apple, Tesla, IBM, Amazon, and Google, with forecast windows of 4, 6, 8, and 10 days. The best reported accuracy reached 75.38% for Apple. Representation fusion outperformed either visual price input or sentiment input alone in that setting. The result is stock-specific, U.S.-centric, and uses social-media sentiment rather than a firm-linked financial-news archive.

A separate regression-fit study used market-news sentiment as an exogenous factor in opening-price models [32]. One reported nonlinear autoregressive exogenous (NARX) specification achieved an Analysis of Variance (ANOVA) significance level of $p = 0.020$. Sentiment improved model fit in that setting. Regression fit is not the same as robust directional forecasting, because a model can improve opening-price estimation while still failing to discriminate reliably between upward and downward moves.

A deeper fusion system, SA-DLSTM, combined an emotion-enhanced CNN, a denoising autoencoder, and an LSTM for stock-movement prediction on the Hang Seng Index [33]. The system captures three design ideas that recur in later literature: sentiment-strength weighting, feature denoising, and sequence modelling over multivariate financial inputs. The target remains an index-level market series, which avoids the harder problem of article-to-firm relevance matching that is central to the present thesis.

The most relevant benchmark-oriented reference for this thesis is a broad horizon-comparative study [9]. The study evaluated 30 forecasting methods over 67 feature setups and three forecast horizons. The reported conclusion is that no single sentiment configuration dominates all horizons. Sentiment effectiveness is conditional on representation and horizon choice. The same finding reinforces the need for conservative reporting, because a sentiment setup that performs well for one horizon may be statistically unremarkable or weak at another.

Table 3.1 maps the surveyed studies onto five dimensions: market and sample, text input, model family, reported result, and main limitation. Two patterns emerge. First, every study with strong reported numbers also depends on either a large developed-market panel or an index-level target. Neither condition holds for bilingual firm-level EGX forecasting. Second, the horizon-comparative benchmark argues that no sentiment configuration dominates across horizons. These two observations motivate the methodological choices in Chapter 4, where bilingual routing, conservative lag, and horizon-specific reporting are treated as design constraints rather than refinements.

3.4 Why Sentiment Does Not Always Improve Forecasting

The surveyed literature does not support a simple claim that sentiment always improves predictive accuracy. Four failure modes recur.

Table 3.1: Quantitative comparison of representative sentiment-based stock forecasting studies.

Study	Market and sample	Text input	Model family	Reported result	Main limitation
DJIA mood study [7]	DJIA, about 9.8 million tweets, Feb.–Nov. 2008	Twitter mood streams	GPOMS mood features with SOFNN	87.6% direction accuracy	Index-level setup with no firm-level entity mapping
Message-board topic study [18]	18 stocks, 1 year, 171 train dates and 78 test dates	Message-board topic sentiment	Topic-sentiment features with linear SVM	+2.07 percentage points mean accuracy over price-only; +9.83 on difficult stocks	Short horizon, shallow topic model, single-language data
S&P 500 panel study [23]	S&P 500 firms, 5 years, more than 265,000 news articles	Financial news	Deep text, price, and fusion models	Large-scale comparison across price-only, text-only, and fusion settings	Developed-market large-cap panel with no bilingual routing problem
LSTM vs. GRU sentiment study [8]	Developed-market stock data with matched financial news	Financial news plus stock features	LSTM and GRU	Sentiment-augmented models improved RMSE, MAPE, and R over stock-only variants	Regression emphasis, no firm-level bilingual challenge
Candlestick fusion study [31]	Five U.S. stocks, 4 to 10 day horizons	Social-media sentiment plus candlestick charts	Two-branch CNN	Best reported accuracy 75.38% for Apple	Stock-specific result and social-media input rather than financial-news input
NARX market-news study [32]	Selected stocks with lagged opening-price models	Market news sentiment	Linear and NARX regression	One reported NARX fit significant at $p = 0.020$	Goodness of fit does not guarantee robust directional discrimination
SA-DLSTM fusion study [33]	Hang Seng Index and public opinion streams	Online sentiment plus market variables	ECNN + DAE + LSTM	Higher accuracy than baselines in the integrated framework	Index-level setting avoids article-to-firm relevance matching
Horizon-comparative benchmark [9]	30 methods, 67 feature setups, 3 horizons	Sentiment time series plus prices	Broad benchmark	No single sentiment setup dominated all horizons	Strong evidence that gains are horizon- and setup-specific

The first failure mode is semantic coarseness. A positive article can describe a low-impact corporate event, while a seemingly neutral article can contain material financing or governance information. A market-attitude study argued that sentiment polarity alone does not consistently explain future price changes and that any useful relation is strongly case-specific [34]. Treating polarity as a sufficient event representation is therefore unsafe.

The second failure mode is temporal misalignment. If article sentiment is joined to prices on the wrong day, or if a target window overlaps the feature window, reported performance can be inflated. The horizon-comparative benchmark shows that different sentiment feature setups behave differently across horizons [9]. Loose alignment can therefore distort comparisons by turning methodological noise into apparent predictive signal.

The third failure mode is data scarcity and imbalance. Sparse firm-level article panels create unstable feature histories, and directional labels often lean toward the majority class. A model can then achieve acceptable raw accuracy by overpredicting the dominant

direction [16, 9]. This issue is especially relevant for emerging markets where article volume is much lower than in U.S. large-cap datasets. The combination of sparse text and imbalanced labels makes balanced accuracy and confusion-matrix analysis more informative than headline accuracy alone.

The fourth failure mode is source mismatch. Social media is fast and abundant, but it is noisy and weakly linked to formal firm events. Financial news is cleaner and more structured, but it is slower and less dense. The contrast between the DJIA mood, message-board topic, and multi-signal studies on one side and the S&P 500 panel, LSTM-vs-GRU, and NARX studies on the other shows that sentiment source changes both the signal quality and the methodological difficulty [7, 18, 28, 23, 8, 32]. No source is universally superior. The choice must match the research objective, the market, and the entity-linking assumptions.

3.5 Domain-Specific and Arabic Language Models

The quality of the sentiment signal depends heavily on the language model used to produce it. Generic sentiment tools often misread financial language because words such as “liability”, “dilution”, or “guidance” carry domain-specific meanings. FinBERT was designed precisely for this issue. The original paper reports a 14 percentage point improvement over the previous state of the art on Financial PhraseBank, which makes it a much stronger candidate for financial English than a generic sentiment classifier [10]. Weak sentence-level sentiment estimation propagates directly into weaker daily forecasting features.

The Arabic side requires an additional specialisation step. AraBERT reported state-of-the-art performance on most tested Arabic NLP tasks, with improvements of up to 2.53 F1 points on the downstream benchmarks [11]. ARBERT and MARBERT extended this line further and introduced the ARLUE benchmark, where the best model achieved an aggregate score of 77.40 and outperformed XLM-R Large despite the latter being substantially larger [12]. These results justify routing Arabic text through Arabic-specialised encoders rather than treating Arabic as a secondary case inside a generic multilingual pipeline.

Variant sensitivity is equally important. A CAMeLBERT comparison across five task types and 12 datasets concluded that variant proximity between pre-training data and downstream data matters more than raw pre-training size [27]. This finding is directly relevant for bilingual EGX forecasting. The problem is not only to classify sentiment, but to classify it in a market where Arabic and English articles occupy different stylistic and domain positions. A single shared model is simpler to deploy but comes at the cost of linguistic fit.

The Arabic-model literature leaves an open gap. These models are evaluated on Arabic NLP benchmarks, not on bilingual financial forecasting benchmarks for the Egyptian market. Strong benchmark performance justifies their use as sentiment encoders. It does not by itself prove that they improve stock prediction. That remaining empirical question is one of the contributions of the present thesis [11, 12, 27].

Recent Arabic financial NLP resources make this gap more visible. The SAHM benchmark introduces Arabic financial and Shari’ah-compliant reasoning tasks [35]. Recent

Saudi-market work studies Arabic financial sentiment with large language models [36]. Both resources support the need for Arabic finance-specific evaluation rather than reliance on general Arabic sentiment benchmarks. Both also clarify a limitation of the present thesis: without independent human-labeled Arabic financial sentiment examples, AraBERT can be adapted to financial text through Masked Language Modeling (MLM), but it cannot be validly evaluated as a supervised Arabic FinBERT equivalent.

3.6 Evaluation Metrics and Research Gap

Comparison across prior forecasting studies is complicated by inconsistent evaluation practice. Regression-oriented papers often report RMSE, MAE, MAPE, or correlation. Directional papers more often report accuracy, precision, recall, and F1. A few papers add simulated returns, which is useful but depends strongly on trading assumptions. These metrics are not interchangeable. A classifier can show a favourable raw accuracy while failing to recover the minority class. A regression model can reduce RMSE while still predicting the sign of the move poorly. Metrics must therefore be interpreted in relation to the target definition and the class balance [8, 9, 16, 33].

The gap this thesis addresses is both geographic and methodological. Most sentiment-based forecasting studies focus on English text, developed markets, single-language pipelines, or index-level prediction. Very few studies examine bilingual Arabic-English financial-news routing for Egyptian equities. Fewer still evaluate that problem under conservative walk-forward validation with explicit leakage and overfitting checks. The present work narrows the scope to three EGX tickers and a single bilingual financial-news source so that article relevance, language routing, and temporal alignment can be studied under cleaner conditions. The contribution is not the assumption that sentiment is universally powerful. The contribution is a test, under a more locally relevant and more tightly controlled design, of how much forecasting value bilingual financial-news sentiment actually carries.

3.7 Chapter Summary

This chapter reviewed the state of the art in sentiment-based stock forecasting and identified the lessons that shape the present thesis. Text helps, but reported gains depend on source quality, feature design, horizon choice, and validation rigour. Financial-news studies are cleaner than social-media studies, yet they remain concentrated in English and in developed markets. Domain-specific financial English models and Arabic-specialised transformers offer strong upstream sentiment encoders, but they were rarely evaluated together in a bilingual EGX forecasting setting. These gaps shaped the methodology in the next chapter.

Chapter 4

Methodology

This chapter explains how the thesis is designed and what was built. It starts with the overall research design and the pipeline, then covers the data sources, article preparation, bilingual sentiment scoring, the formal confidence filter and daily aggregation rules, event-aware variants, target construction, the forecasting setup, evaluation metrics, leakage control, and the four auxiliary experiments. Numbered equations are used wherever the pipeline behaviour needs to be pinned down precisely, and each one is referenced in the text around it.

4.1 Research Design

This thesis is carried out as an industrial bachelor project between the German University in Cairo and Robin, the data-science arm of Beltone Holding. The original project brief asks whether sentiment information provides additional predictive value beyond historical market data in financial forecasting. The practical work evolves in two stages.

The first stage extends an existing EGX direction-forecasting system that already uses historical prices, engineered technical indicators, and machine-learning models. In exploratory development, the addition of a sentiment signal appears to improve directional accuracy by a small absolute margin. That result is useful as motivation, but it does not isolate the value of sentiment itself because the underlying price-based model is already strong and heavily feature-engineered.

The second stage, which forms the core reproducible contribution of this thesis, reframes the problem as a sentiment-only benchmark. The goal is a narrower and more defensible question: if price-derived predictors are removed, how much information does bilingual financial-news sentiment alone provide about future stock direction and return? This design makes the interpretation cleaner. If the sentiment-only benchmark is weak, the study can report that result directly. If it is moderately useful, later integration with price-based systems becomes easier to justify.

The final methodology evaluates three complementary ideas. The first is bilingual sentiment extraction from Arabic and English financial news. The second is leakage-safe alignment of article-derived features to later trading dates. The third is the comparison of all-news, event-only, and event-weighted sentiment representations. Figure 4.1 summarises the end-to-end workflow adopted in the thesis, from bilingual news collection to leakage-safe evaluation.

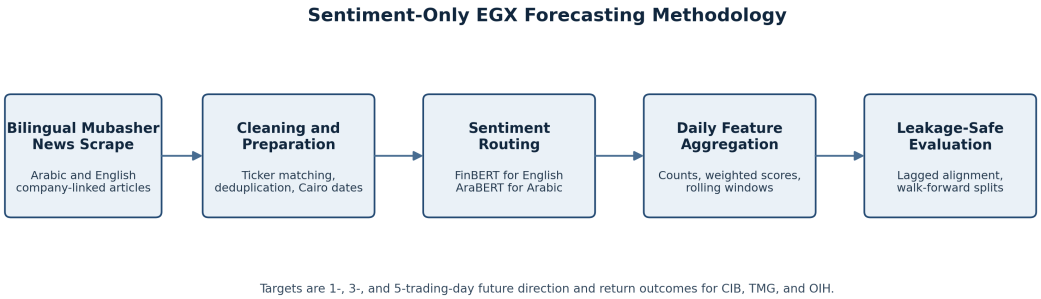


Figure 4.1: End-to-end methodology for the sentiment-only EGX forecasting benchmark.

4.2 Study Universe and Data Sources

The empirical study focuses on three Egyptian Exchange tickers: Commercial International Bank, also written as CIB or COMI; Talaat Moustafa Group, also written as TMG or TMGH; and Orascom Investment Holding, also written as OIH. These tickers are selected because they are actively discussed in local financial coverage and provide a mix of relatively higher-coverage and lower-coverage cases.

The final reproducible benchmark uses Mubasher as the sole news source because it offers both Arabic and English listing pages tied to specific EGX company pages. Restricting the thesis dataset to one source keeps crawling behaviour, ticker linkage, and article structure consistent across the study. Historical market prices are loaded from local CSV files stored in the project repository, with external download used only as a fallback if a local file is missing.

The raw scrape and the cleaned article counts differ because the final dataset applies confidence filtering, text cleaning, language normalisation, and duplicate removal. Table 4.1 summarises the final bilingual article coverage used in the thesis.

Table 4.1: Raw scraped records and final sentiment-labeled article counts used in the thesis benchmark.

Ticker	Raw records	Final articles	English	Arabic
CIB	1064	1013	310	703
TMG	616	604	189	415
OIH	699	304	87	217
Total	2379	1921	586	1335

The same distribution is visualised in Figure 4.2. The figure shows the language imbalance for all three tickers and confirms that OIH has the smallest total article count overall. This asymmetry matters because a sparse text history makes any sentiment-derived predictor less stable and increases the variance of the final estimates.

4.3 News Collection and Article Preparation

The news pipeline begins by scraping Mubasher listing pages separately for Arabic and English. For each ticker-language pair, the crawler collects article identifiers, URLs,

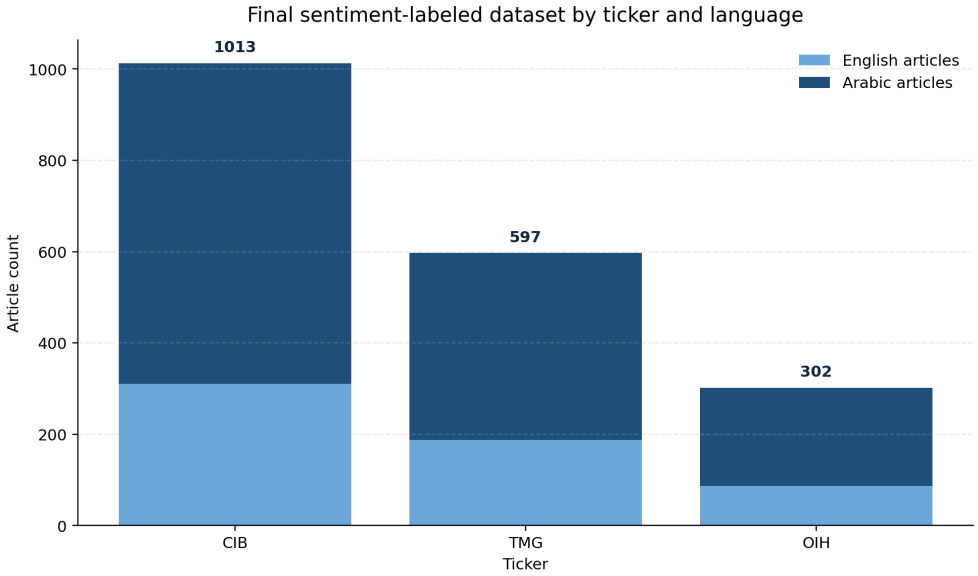


Figure 4.2: Final sentiment-labeled article counts by ticker and language.

titles, summaries, and article bodies. The crawler is incremental and retry-aware so that interrupted collection can be resumed without restarting the archive.

Collected articles pass through a preparation stage that standardises company references and removes low-quality or duplicated records. The preparation stage maps all company name variants to a single reference identity, normalises timestamps to Cairo local time, cleans Arabic and English text, removes very short or duplicated records, and filters weak entity matches. The result is a sentiment-ready bilingual article table whose rows correspond to distinct company-linked news items.

Article relevance is then represented explicitly rather than assumed. A direct article contains the company alias in the title. A contextual article contains the alias in the body but not in the title. An indirect article is linked through weaker alias evidence. A stock-page-only article appears on the company page without stronger textual support. These categories are converted into weights so that strong company references contribute more than weak ones when daily sentiment features are aggregated.

4.4 Bilingual Sentiment Extraction

After article preparation, sentiment is scored using separate models for English and Arabic. English articles are routed to FinBERT [10], while Arabic articles are routed to an AraBERT-based sentiment route [11]. The reason for routing by language instead of forcing both languages through one shared model is methodological. The literature reviewed in Chapter 3 shows that domain-specific financial English models and Arabic-specialised language models are both more defensible than generic multilingual sentiment tools for their respective settings.

For each article, the sentiment layer stores the detected language, the model route, the predicted label, the class probabilities, a model-confidence estimate, and a continuous

sentiment score. The article-level sentiment contribution is then weighted by both ticker-match confidence and model confidence. This design downweights weak ticker matches and low-confidence predictions before any daily aggregation takes place.

Using off-the-shelf transformer models is a practical choice, not a claim that they are the best fit for Egyptian financial language. The goal is to find out whether a bilingual sentiment signal exists at all under a tight benchmark. If it does, later work can fine-tune these models on Arabic and English financial corpora.

4.5 Confidence Filtering and Daily Sentiment Aggregation

Article-level predictions are converted into time-series features through two formal steps: an entity-match confidence filter, then a confidence-weighted daily aggregation.

Entity-match confidence filter. Each candidate article i carries a categorical ticker-match confidence $c_i \in \{\text{low}, \text{medium}, \text{high}\}$ produced by the matcher in `src/news_pipeline`. Let $\text{rank}(\cdot)$ assign $\text{rank}(\text{low}) = 0$, $\text{rank}(\text{medium}) = 1$, $\text{rank}(\text{high}) = 2$. Article i is retained for daily aggregation if and only if

$$\text{rank}(c_i) \geq \text{rank}(\tau), \quad \tau = \text{medium}, \tag{4.1}$$

where the threshold τ is taken from the saved benchmark configuration[1]. Equivalent to (4.1), low-confidence matches are dropped before any sentiment score is aggregated.

Article weighting. Each retained article i contributes a non-negative aggregation weight

$$w_i = w^{\text{rel}}_{r(i)} \cdot w^{\text{match}}_{c_i} \cdot p_i \cdot \mathbf{1}[\text{is_scored}_i = 1], \tag{4.2}$$

where $r(i) \in \{\text{direct}, \text{contextual}, \text{indirect}, \text{stock_page_only}\}$ is the entity-relevance label of article i and $p_i \in [0, 1]$ is the model confidence for the predicted sentiment class. The fixed relevance and match weights are $w^{\text{rel}} = (1.00, 0.85, 0.65, 0.45)$ and $w^{\text{match}} = (0.60, 0.80, 1.00)$ for $c_i \in (\text{low}, \text{medium}, \text{high})$. After the filter in (4.1), only $c_i \in \{\text{medium}, \text{high}\}$ contribute to (4.2).

Article sign. The article-level sign s_i derives from the predicted sentiment label ℓ_i as

$$s_i = \begin{cases} +1, & \ell_i = \text{positive} \\ -1, & \ell_i = \text{negative} \\ 0, & \ell_i = \text{neutral} \end{cases} \tag{4.3}$$

[1] Default `min_match_confidence` in `SentimentOnlyForecastConfig` inside `src/egx_direction/sentiment_only_pipeline.py`.

Daily weighted sentiment. For ticker s and source calendar day $d-1$, the daily weighted sentiment is

$$S_{s,d} = \begin{cases} \dfrac{\sum_{i \in \mathcal{A}_{s,d-1}} s_i \cdot w_i}{\sum_{i \in \mathcal{A}_{s,d-1}} w_i}, & \text{if } \sum_{i \in \mathcal{A}_{s,d-1}} w_i > 0 \\ 0, & \text{otherwise (missingness flag set to 1)} \end{cases} \tag{4.4}$$

where $\mathcal{A}_{s,d-1}$ is the set of retained articles linked to ticker s whose Cairo-local publication date equals $d-1$. The window therefore lags the trading date d by one calendar day so that no same-day article is seen at prediction time. This corresponds to the configured value `sentiment_lag_calendar_days` $= 1$ in the saved `run_config.json` of the sentiment-stock-impact study and to `lag_strategy=previous_day` (one-day shift) in the news-pipeline aggregator.

Daily missingness handling. Equation (4.4) sets $S_{s,d} = 0$ when the denominator vanishes. The pipeline also records a binary *article_has_news* indicator so that downstream models can distinguish a genuine zero-sentiment reading from a no-article day. Rolling windows and recency indicators derived later in the pipeline are computed on the daily series obtained from (4.4) after the one-day shift in (4.2) and (4.4).

Three language views are retained throughout the experiments: an English-only view ($\mathcal{A}^{\text{en}}_{s,d-1}$), an Arabic-only view ($\mathcal{A}^{\text{ar}}_{s,d-1}$), and a bilingual view that combines both languages after routing and scoring. This separation matters because the Egyptian information environment is genuinely bilingual. The experiments therefore test whether combining both languages adds signal or whether it adds noise.

4.6 Event-Aware Feature Variants

In addition to standard daily sentiment aggregation, the thesis evaluates event-aware features. Articles differ not only in polarity but also in event type, and not all events are equally likely to move a stock. The final pipeline labels articles using broad market-moving categories such as earnings, dividends, mergers and acquisitions, contracts and projects, expansion and investment, financing, and governance or disclosure.

These labels support three feature strategies. The all-news strategy keeps every retained article in the daily aggregation in (4.4). The event-only strategy restricts the aggregation set $\mathcal{A}_{s,d-1}$ to event-related articles. The event-weighted strategy keeps all retained articles but multiplies the weight in (4.2) by an additional event factor for event-related rows. This is not a full thematic NLP model, but it is a practical intermediate step toward richer event-aware relevance modelling.

4.7 Target Construction and Lagged Alignment

The target variables derive from future closing prices. For each trading day t and forecast horizon h, the future return is defined as

$$r_{t,h} = \frac{P_{t+h}}{P_t} - 1, \tag{4.5}$$

where P_t is the closing price on day t and P_{t+h} is the closing price after h future trading days. A binary direction target is then defined as

$$y_{t,h}^{\text{dir}} = \begin{cases} 1, & r_{t,h} > 0 \\ 0, & r_{t,h} \leq 0 \end{cases} \tag{4.6}$$

The thesis evaluates three short forecast horizons, reported with the corresponding implementation settings in Chapter 5.

Temporal alignment is one of the most important parts of the methodology. The daily sentiment $S_{s,d}$ in (4.4) is constructed from the prior calendar day, so the features available for trading day t contain only news published before that trading date. This conservative lag is essential for avoiding subtle look-ahead effects and matches the saved configuration constants quoted above.

4.8 Forecasting Framework

The forecasting layer is intentionally simple because the objective is to isolate the informational value of sentiment rather than to maximise performance through model complexity. The thesis therefore uses one low-capacity classifier for direction and one low-capacity regressor for future return. Feature standardisation, model selection, and classification-threshold selection are all performed chronologically on validation data only.

Each ticker panel is sorted by date and divided into chronological training, validation, and test regions. Adjacent regions are separated by an embargo at least as long as the forecast horizon, which prevents overlapping target windows from leaking across splits. Final predictions are generated in an expanding-window walk-forward scheme. At every step, the model is trained only on rows whose target values are already known at the prediction date. This procedure is slower than a random split, but it is more credible. The exact hyperparameter values appear in Chapter 5 together with the experimental setup.

4.9 Evaluation Metrics

The study reports both classification and regression metrics because the task has two outputs: future direction and future return. Directional evaluation uses accuracy, balanced accuracy, precision, recall, F1, and confusion-matrix counts. Return evaluation uses mean absolute error, root mean squared error, return sign accuracy, and the correlation between predicted and realised returns.

Balanced accuracy is treated as a primary metric rather than as an auxiliary one because the direction labels are not expected to be perfectly balanced across all ticker-horizon combinations. A model that leans too strongly toward the majority direction can still appear acceptable under raw accuracy alone. The reported metric set is therefore designed to expose that behaviour rather than to conceal it.

4.10 Leakage Control and Validity Checks

Leakage is a common failure mode in financial forecasting, so the final pipeline includes explicit validity checks beyond the main experiment loop. These checks verify that the historical rows used for any validation or test prediction do not extend into the future target window of the evaluation row. The checks also inspect source coverage and compare training, validation, and test behaviour, which prevents acceptance of suspiciously optimistic results without further scrutiny.

The overall intention of the methodology is conservative. The thesis prefers a weaker but believable result over an inflated result produced by loose alignment, optimistic evaluation, or metric choice. That choice matters even more given how thin bilingual EGX data is.

4.11 Auxiliary Experiment Protocols

Alongside the main benchmark, the thesis runs four auxiliary experiments to show how the research evolved and where it should go next. They do not replace the accepted final benchmark. Their job is to test specific surrounding claims and to make sure no exploratory work gets reported without proper documentation.

The first auxiliary experiment returns to the original industrial motivation. An external EGX direction-forecasting repository developed during the Robin/Beltone internship is re-run as a historical price-based baseline. The same chronological train-test split is used for each ticker-model pair, and the baseline feature set is compared against an augmented version that adds one-day-lagged bilingual sentiment features. This design tests whether sentiment adds value inside a richer market-feature model, but it is treated as exploratory because the target formulation and feature regime differ from the stricter sentiment-only benchmark.

The second auxiliary experiment evaluates direct sentiment-return association. Daily bilingual sentiment is shifted by one calendar day, aligned to later trading dates, and correlated with same-day, 1-day, 3-day, and 5-day returns. Pearson and Spearman correlations are reported together with five-day block-permutation p-values. This experiment is descriptive rather than causal. Its purpose is to check whether the raw sentiment score has a stable monotonic relationship with later returns before more complex forecasting models are interpreted.

The third auxiliary experiment tests Arabic financial-domain adaptation. The Arabic article corpus is used for Masked Language Modeling adaptation of AraBERT, following the first stage of the FinBERT-style domain-adaptation logic. The adapted checkpoint is then evaluated against the original AraBERT model on held-out Arabic financial text under identical conditions: the same chronological 80/20 validation split, the same fixed seed of 42 across Python, NumPy, and PyTorch, the same base AraBERT tokenizer for both runs, and the identical evaluation script. This is a language-modelling evaluation, not a sentiment classification benchmark, and is therefore reported as a domain language modelling fit measurement rather than as supervised sentiment performance. A supervised Arabic financial sentiment model requires independent human-labeled examples, which are not available in the current thesis dataset.

The fourth auxiliary experiment evaluates social-media feasibility. The target is a free, ethical, and officially approved route for adding social-media data. Reddit and X/Twitter are considered as candidate sources. The protocol rejects unofficial scraping and paid-access bypasses. Social media is therefore kept as future work unless approved official access is obtained and the resulting data can be handled without redistributing raw user content.

4.12 Chapter Summary

This chapter described the full solution design. It starts from bilingual Mubasher news, scores articles using language-specific sentiment models, applies the confidence filter and weighted aggregation defined in equations (4.1)–(4.4), shifts the resulting daily features back by one calendar day to avoid look-ahead, and evaluates them under walk-forward validation. The four auxiliary experiment protocols were also defined here. The next chapter reports the implementation settings and the results.

Chapter 5

Results & Implementation

This chapter reports the implementation settings and the empirical results. It opens with a reproducibility table, then covers the final dataset, the main multi-horizon results, the event-filtering results, the validity checks, and the four auxiliary experiments. A discussion section at the end explains what the findings mean.

5.1 Experimental Setup

Table 5.1 records every parameter needed to reproduce any number reported in this chapter. The values are drawn directly from the project `requirements.txt`, `pyproject.toml`, the AraBERT training script, the AraBERT evaluation script, and the saved `run_metadata.json` files under `outputs/`.

Table 5.1: Experimental setup parameters used for every reported experiment.

Parameter	Value
Python version	3.9.6
Operating system	macOS 26.5 Tahoe (Darwin 25.5.0, arm64); Apple macOS 26 uses Darwin 25.x (kernel version lags product version by one)
CPU / Apple Silicon	Apple M4
System memory	16 GB unified memory
GPU / VRAM	None (Apple Silicon Metal Performance Shaders available; final AraBERT runs executed on CPU device per saved `run_metadata.json`)
CUDA / cuDNN	Not applicable (no NVIDIA GPU used)
PyTorch version	2.8.0
Transformers version	4.57.3
Scikit-learn version	1.6.1
Pandas version	2.3.3
NumPy version	1.26.4

Continued on next page

Table 5.1 – continued from previous page

Parameter	Value
Datasets library	Not used (PyTorch `DataLoader` with custom `TextDataset`)
AraBERT base checkpoint	`aubmindlab/bert-base-arabertv02`
AraBERT base revision (HuggingFace SHA)	`016fb9d6768f522a59c6e0d2d5d5d43a4e1bff60`
Random seed (Python, NumPy, PyTorch, `torch.cuda`)	42
Deterministic flags	`torch.manual_seed(42)`, `np.random.seed(42)`, `torch.cuda.manual_seed_all(42)` (no GPU active); `torch.use_deterministic_algorithms(True, warn_only=True)` set in all training and evaluation scripts (`warn_only=True` because Apple MPS does not support all deterministic ops); `CUBLAS_WORKSPACE_CONFIG=:4096:8` set via `os.environ`; `torch.backends.cudnn.benchmark` not applicable (no CUDA)
AraBERT MLM training: optimiser	AdamW
AraBERT MLM training: learning rate	5×10^{-5}
AraBERT MLM training: warmup	None (constant learning rate)
AraBERT MLM training: weight decay	0 (PyTorch `AdamW` default unmodified)
AraBERT MLM training: epochs	1.0 (capped by `max-steps`)
AraBERT MLM training: max steps	20
AraBERT MLM training: batch size	2
AraBERT MLM training: max sequence length	256
AraBERT MLM training: MLM probability	0.15
AraBERT MLM evaluation: batch size	4
AraBERT MLM evaluation: max sequence length	256
AraBERT MLM evaluation: MLM probability	0.15
LoRA / quantization	Not used in any reported run
Sentiment-only direction model	`sklearn` logistic regression with balanced class weights

Continued on next page

Table 5.1 – continued from previous page

Parameter	Value
Sentiment-only return model	`sklearn` ridge regression
Logistic regularisation grid (C)	$\{0.05, 0.1, 0.5\}$, selected on validation
Ridge regularisation (α)	10.0
Minimum training history	250 rows
Walk-forward refit cadence	every 250 evaluation rows
Embargo between adjacent splits	at least the forecast horizon (1, 3, or 5 trading days)
Sentiment lag (calendar days)	1 (per equation (4.4) in Chapter 4)
Entity-match confidence threshold τ	medium (per equation (4.1))

Table 5.1 fixes every numerical and software-version parameter that affects an output. All scripts, data manifests, evaluation CSVs, and audit reports are released through the project repository at https://github.com/hussienhaithamm/EGX-Sentiment-Forecasting-Thesis so that every reported number can be regenerated from the same configuration.

5.2 Final Experimental Dataset

The final sentiment-ready dataset contains 1,921 cleaned and sentiment-labeled Mubasher articles across the three target tickers. Event labelling identifies 1,730 of these articles as belonging to one of the event categories used in the event-aware experiments. Table 5.2 summarises the ticker-level counts used in the final implementation.

Table 5.2: Final cleaned dataset and event-labeled subset used in the reported experiments.

Ticker	Final articles	Event-labeled articles
CIB	1013	874
TMG	604	570
OIH	304	286
Total	1921	1730

The event counts are high relative to total article counts, which indicates that Mubasher company pages are strongly event driven rather than dominated by generic commentary. Coverage remains uneven across tickers. CIB has a much denser article history than OIH, and this asymmetry matters when the results are interpreted per ticker rather than only in aggregate.

5.3 Main Multi-Horizon Results

The final accepted run evaluates three forecast horizons: 1, 3, and 5 trading days. Direction prediction uses logistic regression with balanced class weights, while return prediction

uses ridge regression. Features are standardised before model fitting. The classification regularisation parameter is tuned over the grid $\{0.05, 0.1, 0.5\}$. The ridge configuration uses $\alpha = 10.0$. Classification thresholds are not fixed at 0.5. They are selected from validation predictions only and applied unchanged to the test block. The chronological setup matches Chapter 4, while the implementation adds the concrete numerical settings already listed in Table 5.1.

Table 5.3 reports the final all-news bilingual results that passed the chronological evaluation pipeline. The table includes the summary metrics most relevant to the thesis claims: raw accuracy, balanced accuracy, F1, return sign accuracy, and the correlation between predicted and realised future returns.

Table 5.3: Final all-news bilingual sentiment-only results across the three forecast horizons.

Horizon	Ticker	Accuracy	Balanced Acc.	F1	Sign Acc.	Return Corr.
1	CIB	0.487	0.501	0.264	0.476	-0.026
1	TMG	0.494	0.506	0.637	0.451	-0.028
1	OIH	0.449	0.492	0.588	0.447	0.065
3	CIB	0.510	0.479	0.661	0.490	0.002
3	TMG	0.499	0.503	0.573	0.480	-0.104
3	OIH	0.467	0.505	0.030	0.539	-0.037
5	CIB	0.449	0.500	0.141	0.531	-0.022
5	TMG	0.489	0.486	0.650	0.534	-0.020
5	OIH	0.425	0.503	0.021	0.585	0.026

Table 5.3 shows a weak and inconsistent sentiment-only signal. The best accepted ticker-horizon pair is OIH at the 5-day horizon, with test accuracy 0.425, balanced accuracy 0.503, F1 0.021, sign accuracy 0.585, and return correlation 0.026. The best balanced accuracy in the final accepted run appears for TMG at the 1-day horizon, where balanced accuracy reaches 0.506. These values are above random only marginally. They do not support a claim that bilingual news sentiment alone is a strong standalone EGX forecasting model.

Another pattern is that F1 can look stronger than balanced accuracy. This occurs when the model predicts the majority upward class more often than the minority downward class. Several rows show F1 above 0.60 while balanced accuracy remains close to 0.50. Balanced accuracy is therefore more informative than raw accuracy or F1 alone in an imbalanced directional setting.

5.4 Event Filtering and Event Weighting

The thesis also evaluates whether event filtering improves the signal. On aggregate, it marginally does. Relative to the all-news bilingual baseline, the mean event-only accuracy change is +0.025 and the mean event-weighted accuracy change is +0.032. Table 5.4 summarises these aggregate comparisons.

The event-aware variants produce a few locally interesting cases. CIB improves from 0.487 to 0.499 accuracy at the 1-day horizon under the event-only setting, and from 0.510 to 0.525 at the 3-day horizon. OIH at the 5-day horizon improves from 0.425 to 0.563 accuracy and from 0.021 to 0.700 F1 under event weighting. These gains are driven by a

Table 5.4: Aggregate event-filtering effects relative to the all-news bilingual baseline.

Strategy	Mean accuracy delta vs. all-news	Best positive delta
Event-only	+0.025	+0.117
Event-weighted	+0.032	+0.138

small number of large improvements: 7 of 9 ticker-horizon pairs show positive accuracy deltas under both strategies, but the magnitude is highly concentrated in a few cases. The broader conclusion is that event information helps selectively, and the aggregate positive mean is carried largely by OIH at the 5-day horizon, which also shows the highest overfitting risk.

5.5 Validity Checks and Accepted Final Model

The strongest raw-accuracy candidate observed in the final multi-horizon event run was OIH at the 5-day horizon under the event-weighted bilingual setting. Test accuracy reached 0.563 and F1 reached 0.700. The corresponding row was flagged by the Phase 7 validity analysis because training accuracy rose to 0.609 while validation accuracy fell to 0.485. The result is therefore exploratory rather than accepted.

The accepted final candidate is OIH at the 5-day horizon under the all-news bilingual setting. Its validation accuracy, balanced accuracy, and F1 are 0.524, 0.501, and 0.057. Its test accuracy, balanced accuracy, F1, and sign accuracy are 0.425, 0.503, 0.021, and 0.585. This candidate is retained because it passes the leakage and overfitting filters and its validation balanced accuracy clears the minimum acceptance threshold of 0.490. The near-zero F1 on both validation and test indicates that the logistic regression threshold selected from validation predictions pushes the classifier heavily toward one class; sign accuracy (0.585), which does not depend on the classification threshold, remains the more informative metric for this candidate.

The Phase 7 validity results quantify the reliability of the full pipeline. Leakage checks pass for 45 panel evaluations covering 43,880 validation and test prediction rows, and no history violations are detected. Coverage analysis shows that TMG and OIH match the currently available Mubasher listing pages, while CIB has one additional Arabic listing page available at audit time. Overfitting analysis flags 14 of the 45 experiment rows, which confirms that raw test accuracy alone would overstate the quality of the strongest-looking candidate.

Coverage remains a substantive limitation. Prior-7-day article coverage is 0.569 for CIB, 0.387 for TMG, and 0.307 for OIH. These values explain part of the instability seen in the ticker-level results. The weakest panels are not simply weaker models. They are weaker information histories.

5.6 Auxiliary Experiments and Research Journey

The final sentiment-only benchmark was not the first experimental form of the project. The research process began with the industrial question of whether a sentiment signal

improves an existing EGX direction-forecasting model based on historical prices and technical features. That historical integration experiment uses the Robin/Beltone internship codebase as an external baseline and compares the original historical features against the same feature set augmented with one-day-lagged bilingual sentiment features. The experiment runs for CIB, TMG, and OIH using logistic regression, XGBoost, and a Multilayer Perceptron (MLP) under a shared chronological test period.

Table 5.5: Summary of auxiliary experiments added to document the research journey.

Experiment	Purpose	Main outcome
Historical model integration	Test sentiment as an added feature in the Robin/Beltone EGX historical forecasting model	Sentiment improves raw accuracy in 5 of 9 ticker-model comparisons; gains are conditional rather than uniform
Sentiment-return association	Check whether lagged daily sentiment correlates directly with stock returns	No main lagged weighted-sentiment feature meets the pre-specified Spearman and permutation threshold
Leakage and overfitting audit	Verify that final results are not inflated by future information or optimistic model selection	Leakage status passes across 45 panel checks and 43,880 validation/test rows; 14 of 45 rows are flagged for overfitting
AraBERT financial adaptation	Test whether Arabic financial news improves AraBERT masked-language modelling	Held-out MLM loss decreases from 0.9414 to 0.8238 relative to base AraBERT (measures domain language modelling fit, not supervised sentiment performance, which remains future work pending Arabic financial sentiment labels)
Social-media feasibility	Determine whether social data can be added through free, ethical, official access	Reddit requires approved official access; X/Twitter broad public search is not free under the official API model

The historical integration result is mixed. Sentiment improves raw test accuracy in 5 of 9 ticker-model comparisons. The largest gain appears for OIH with MLP: accuracy increases by 0.143, balanced accuracy by 0.106, and F1 by 0.309. The largest loss appears for CIB with MLP: accuracy decreases by 0.105. Logistic regression is mostly flat or weaker after sentiment is added. XGBoost shows small gains for CIB and OIH and a small loss for TMG. The result is useful as motivation but does not cleanly isolate the value of sentiment because the baseline already contains many price-derived predictors. That limitation is the reason the thesis moves toward the stricter sentiment-only benchmark reported above.

A second auxiliary experiment inspects the raw relationship between daily news sentiment and stock returns. Daily bilingual sentiment is shifted by one calendar day before alignment to trading dates, then correlated with same-day, 1-day, 3-day, and 5-day returns. Pearson and Spearman correlations are computed, and p-values are estimated by block permutation with five-day blocks. No ticker-horizon pair for the main lagged weighted-sentiment feature meets the pre-specified threshold of absolute Spearman correlation at least 0.10 and permutation p-value at most 0.05. The strongest displayed associations remain weak: OIH 3-day Spearman correlation is -0.127 with p-value 0.155, OIH 5-day is -0.109 with p-value 0.186, and TMG same-day is -0.102 with p-value 0.120. This supports a conservative interpretation. Sentiment shows weak descriptive associations but not a stable monotonic return relation by itself.

The Arabic sentiment component is also tested from a domain-adaptation perspective. Following the logic of FinBERT, a valid Arabic financial analogue requires finance-domain language-model adaptation followed by supervised sentiment fine-tuning on independent human-labeled Arabic financial examples. The available thesis corpus supports only the

first stage. The AraBERT domain-adaptation run used 1,327 Arabic financial articles available at training time, split chronologically into 1,061 Masked Language Modeling (MLM) training articles and 266 validation articles. A short CPU-based AraBERT adaptation run completes for 20 steps, with training loss 0.9676 and validation loss 0.8218 measured during training. A follow-up fixed-mask evaluation compares the adapted checkpoint against the original AraBERT under identical conditions: the same chronological 80/20 validation split, the same fixed seed of 42 across Python, NumPy, and PyTorch, the same base AraBERT tokenizer for both runs, and the identical evaluation script. Table 5.6 reports the resulting per-split MLM loss and perplexity on held-out Arabic financial news text. This measures domain language modelling fit, not supervised sentiment performance, which remains future work pending Arabic financial sentiment labels. Supervised Arabic financial sentiment fine-tuning is excluded because the present labels are model-generated outputs from the current AraBERT route and do not constitute independent human annotations [10, 11].

Table 5.6: Masked-language-model evaluation of base AraBERT and the financial-domain adapted checkpoint. Both rows of each split use the same chronological 80/20 validation split, the same fixed seed of 42 (Python, NumPy, PyTorch), the same base AraBERT tokenizer, and the identical evaluation script `scripts/evaluate_arabert_financial_adaptation.py`. The reported values measure domain language modelling fit, not supervised sentiment performance, which remains future work pending Arabic financial sentiment labels.

Split	Model	Examples	Masked tokens	Loss	Perplexity
Train	Base AraBERT	1061	40403	0.9131	2.492
Train	Financial-adapted AraBERT	1061	40403	0.7882	2.199
Validation	Base AraBERT	266	10080	0.9414	2.564
Validation	Financial-adapted AraBERT	266	10080	0.8238	2.279

Validation MLM loss on held-out Arabic financial text falls from 0.9414 to 0.8238, while perplexity falls from 2.564 to 2.279, corresponding to a 12.49% relative loss reduction. This supports the narrower claim that the adapted checkpoint models Arabic financial text better than the base checkpoint under the MLM objective on the same evaluation script and seed. This measures domain language modelling fit, not supervised sentiment performance, which remains future work pending Arabic financial sentiment labels.

The final auxiliary investigation considers social-media sources. The requirement is a free, ethical, and officially approved route for EGX social-media sentiment. The conclusion is restrictive. Reddit is considered only through official API or approved research access and with limits on redistribution and model training. X/Twitter is excluded from the current implementation path because broad public-read access is paid under the official API model, while unofficial scraping is inappropriate for an academic thesis. Prior work on Arabic tweets and Arab stock-market indices, including EGX30, supports the relevance of this direction without removing the current access constraint [30].

5.7 Interpretation

The final evidence supports a limited rather than optimistic conclusion. Bilingual news sentiment contains some directional information. The signal is not stable enough to serve

as a strong standalone EGX forecasting system in the present dataset. The strongest accepted result is only marginally above random under balanced accuracy. Several apparently strong F1 values are driven by class skew rather than by robust up-versus-down discrimination.

The mixed result is still informative for two reasons. First, it justifies the conservative methodology because looser reporting would have selected the overfit OIH event-weighted model as the headline result. Second, it clarifies where later work should focus: richer relevance modelling, denser news coverage, Arabic financial-domain adaptation with human sentiment labels, official social-media access, and eventual integration with price-derived features rather than sentiment-only forecasting.

5.8 Chapter Summary

This chapter reported the implementation settings and the empirical results. The benchmark uses bilingual Mubasher news, simple walk-forward models with leakage controls, and explicit validity checks. The best accepted result is OIH at the 5-day horizon, but its balanced accuracy stays only slightly above random. Event filtering shows a small positive average gain, though it is concentrated in a few panels. The auxiliary experiments round out the picture: sentiment helps historical models in some cases, the raw sentiment-return correlation is weak, AraBERT domain adaptation lowers held-out MLM loss, and social media remains out of scope without official access. The next chapter draws the overall conclusions and points to future directions.

Chapter 6

Conclusion & Future Work

This chapter answers the three research questions, summarises what was built, and outlines five directions for future work. No new results appear here; every number refers back to Chapter 5.

6.1 Conclusion

This thesis tested whether bilingual financial-news sentiment supports short-horizon EGX stock forecasting under a conservative and reproducible evaluation design. The study began from an industrial motivation: sentiment appeared useful as an additional signal inside an existing historical price-based forecasting model. The re-run of that integration experiment showed conditional gains rather than uniform improvement. Sentiment improved raw accuracy in 5 of 9 ticker-model comparisons, with the strongest gain for OIH using an MLP, but it reduced performance in other settings. The result was useful as motivation rather than a clean final claim. This directly answers RQ3: gains seen under a richer feature setup do not hold up once the evaluation moves to a strict sentiment-only benchmark.

The core contribution of the thesis is the sentiment-only benchmark itself, which addresses RQ1 and RQ2. The implemented pipeline collects Arabic and English Mubasher company news, routes English articles to FinBERT and Arabic articles to an AraBERT-based sentiment route, aggregates article-level sentiment into lagged daily features through the confidence filter and weighted aggregation in equations (4.1)–(4.4), and evaluates all-news, event-only, and event-weighted variants under chronological walk-forward validation. The final accepted result is modest. The strongest validity-passing candidate is OIH at the 5-day horizon under the all-news bilingual setting, with test accuracy 0.425, balanced accuracy 0.503, F1 0.021, and return sign accuracy 0.585. These numbers point to a weak signal rather than a reliable standalone forecasting system. Sign accuracy (0.585) is the more useful metric here because the near-zero F1 reflects the classifier collapsing toward one class after threshold selection, not a complete absence of signal. Event filtering gave a small positive average accuracy gain (+0.025 for event-only, +0.032 for event-weighted), though it was concentrated in a few panels. That is the direct answer to RQ2: event-aware aggregation helps selectively but not consistently.

The mixed results are still scientifically useful. Leakage checks passed across 45 panel evaluations and 43,880 prediction rows, while overfitting analysis flagged 14 of those

45 rows. The strongest raw-accuracy event-weighted candidate was therefore treated as exploratory rather than accepted. A separate correlation study found no stable monotonic link between sentiment and returns. Taken together, these checks support the main conclusion: bilingual financial-news sentiment carries some signal in this EGX setting, but the signal is sparse, unstable, and very sensitive to coverage, horizon, and how strictly the evaluation is run.

6.2 Future Work

The most immediate future step is to improve data coverage. An incremental scrape run at submission time expanded the dataset from 1,912 to 1,921 articles, recovering 17 new raw records (CIB +8, TMG +7, OIH +2) of which 9 passed the medium-confidence entity filter. Prior-7-day article coverage remains 0.569 for CIB, 0.387 for TMG, and 0.307 for OIH, which confirms that sparse text histories limit signal stability for TMG and OIH. Broader source coverage should be considered for future work, but only if ticker linkage and timestamp quality remain auditable. Possible additions include official company disclosures, Egyptian Exchange announcements, and other financial-news providers with stable archives.

A second direction is Arabic financial-domain adaptation. The thesis prepares and tests the first stage of this process by running a short Masked Language Modeling (MLM) adaptation of AraBERT on 1,327 Arabic financial articles. The adapted checkpoint reduces held-out MLM loss relative to base AraBERT under matched seed, tokenizer, and evaluation script. This measures domain language modelling fit, not supervised sentiment performance, which remains future work pending Arabic financial sentiment labels. A valid Arabic FinBERT-style model still requires independent human-labeled Arabic financial sentiment examples for supervised fine-tuning and evaluation. The next step would be to build a small expert-labeled Arabic financial sentiment set with positive, negative, and neutral labels, check inter-annotator agreement, fine-tune the adapted encoder with a classification head, and compare it against the current AraBERT route on held-out human labels. Using model-generated labels as ground truth only copies the existing classifier; it does not show real improvement in sentiment quality.

A third direction is controlled integration with market features. The historical integration experiment suggests that sentiment may help selected model-ticker pairs when added to price-derived features. That experiment should be repeated with a fully unified target definition, walk-forward protocol, and ablation design before making stronger claims. A stronger follow-up compares price-only, sentiment-only, and price-plus-sentiment models under the same target, embargo, validation, threshold-selection, and test-period rules.

A fourth direction is richer event modelling. The event-aware layer can be expanded beyond keyword-style categories. The present event labels are useful as a lightweight relevance signal, but they do not capture event materiality, novelty, or whether the article describes expected or surprising information. A future system trains a supervised event classifier, estimates event severity, and separates company-specific events from market-wide coverage.

The fifth direction is social media. Social media remains future work rather than a completed experiment. Reddit is potentially feasible only through official API or approved

research access with strict data-handling limits. X/Twitter is not a free broad-search option under the current official API model. If approved access becomes available, the next design collects posts only through official channels, avoids redistribution of raw user content, maps Arabic and English aliases for EGX tickers, and compares news-only, social-only, news-plus-social, and ensemble models under the same leakage-safe evaluation framework used in this thesis.

Appendix

Appendix A

Lists

List of Abbreviations

ANOVA Analysis of Variance.

API Application Programming Interface.

AraBERT Arabic Bidirectional Encoder Representations from Transformers.

ARBERT Arabic Bidirectional Encoder Representations from Transformers (Abdul-Mageed variant).

ARLUE Arabic Language Understanding Evaluation benchmark.

AZFinText Arizona Financial Text system.

BERT Bidirectional Encoder Representations from Transformers.

CAMeLBERT Computational Approaches to Modeling Language Bidirectional Encoder Representations from Transformers.

CIB Commercial International Bank.

CNN Convolutional Neural Network.

COMI Commercial International Bank (alternative EGX ticker symbol).

CPU Central Processing Unit.

CSV Comma-Separated Values.

DAE Denoising Autoencoder.

DJIA Dow Jones Industrial Average.

ECNN Emotion-enhanced Convolutional Neural Network.

EGX Egyptian Exchange.

EGX30 Egyptian Exchange 30-stock benchmark index.

FinBERT Financial Bidirectional Encoder Representations from Transformers.

GPOMS Google Profile of Mood States.

GPU Graphics Processing Unit.

GRU Gated Recurrent Unit.

GUC German University in Cairo.

LLM Large Language Model.

LSTM Long Short-Term Memory.

MAE Mean Absolute Error.

MAPE Mean Absolute Percentage Error.

MARBERT Multi-dialect Arabic Bidirectional Encoder Representations from Transformers.

MLM Masked Language Modeling.

MLP Multilayer Perceptron.

NARX Nonlinear Autoregressive Exogenous model.

NLP Natural Language Processing.

OIH Orascom Investment Holding.

RMSE Root Mean Squared Error.

RQ Research Question.

SA-DLSTM Sentiment-Aware Deep Long Short-Term Memory model.

SAHM Arabic Financial and Shari'ah-Compliant Reasoning benchmark.

SOFNN Self-Organizing Fuzzy Neural Network.

SVM Support Vector Machine.

TMG Talaat Moustafa Group.

TMGH Talaat Moustafa Group (alternative EGX ticker symbol).

VRAM Video Random-Access Memory.

XLM-R Cross-lingual Language Model RoBERTa.

List of Figures

4.1 End-to-end methodology for the sentiment-only EGX forecasting benchmark. 18
4.2 Final sentiment-labeled article counts by ticker and language. 19

Bibliography

[1] E. F. Fama, "Efficient capital markets: A review of theory and empirical work," *The Journal of Finance*, vol. 25, no. 2, pp. 383–417, 1970.

[2] A. W. Lo and A. C. MacKinlay, "Stock market prices do not follow random walks: Evidence from a simple specification test," *The Review of Financial Studies*, vol. 1, no. 1, pp. 41–66, 1988.

[3] O. B. Sezer, M. U. Gudelek, and A. M. Ozbayoglu, "Financial time series forecasting with deep learning: A systematic literature review: 2005–2019," *Applied Soft Computing*, vol. 90, p. 106181, 2020.

[4] P. C. Tetlock, "Giving content to investor sentiment: The role of media in the stock market," *The Journal of Finance*, vol. 62, no. 3, pp. 1139–1168, 2007.

[5] C. Kearney and S. Liu, "Textual sentiment in finance: A survey of methods and models," *International Review of Financial Analysis*, vol. 33, pp. 171–185, 2014.

[6] S. R. Das and M. Y. Chen, "Yahoo! for amazon: Sentiment extraction from small talk on the web," *Management Science*, vol. 53, no. 9, pp. 1375–1388, 2007.

[7] J. Bollen, H. Mao, and X. Zeng, "Twitter mood predicts the stock market," *Journal of Computational Science*, vol. 2, no. 1, pp. 1–8, 2011.

[8] K. R. Dahal, N. R. Pokhrel, S. Gaire, S. Mahatara, R. P. Joshi, A. Gupta, H. R. Banjade, and J. Joshi, "A comparative study on effect of news sentiment on stock price prediction with deep learning architecture," *PLOS ONE*, vol. 18, no. 4, p. e0284695, 2023.

[9] C. M. Liapis, A. Karanikola, and S. Kotsiantis, "Investigating deep stock market forecasting with sentiment analysis," *Entropy*, vol. 25, no. 2, p. 219, 2023.

[10] D. Araci, "Finbert: Financial sentiment analysis with pre-trained language models," *arXiv preprint arXiv:1908.10063*, 2019.

[11] W. Antoun, F. Baly, and H. Hajj, "Arabert: Transformer-based model for arabic language understanding," in *OSACT Workshop*, 2020.

[12] M. Abdul-Mageed, A. Elmadany, and E. M. B. Nagoudi, "Arbert & marbert: Deep bidirectional transformers for arabic," *arXiv preprint arXiv:2101.01785*, 2021.

[13] G. S. Atsalakis and K. P. Valavanis, "Surveying stock market forecasting techniques—part ii: Soft computing methods," *Expert Systems with Applications*, vol. 36, no. 3, pp. 5932–5941, 2009.

[14] T. Fischer and C. Krauß, "Deep learning with long short-term memory networks for financial market predictions," *European Journal of Operational Research*, vol. 270, no. 2, pp. 654–669, 2018.

[15] D. M. Q. Nelson, A. C. M. Pereira, and R. A. de Oliveira, "Stock market's price movement prediction with lstm neural networks," in *2017 International Joint Conference on Neural Networks*, pp. 1419–1426, 2017.

[16] Y. Shi, Y. Zheng, K. Guo, and X. Ren, "Stock movement prediction with sentiment analysis based on deep learning networks," *Concurrency and Computation: Practice and Experience*, vol. 33, no. 20, p. e6076, 2021.

[17] T. Loughran and B. McDonald, "When is a liability not a liability? textual analysis, dictionaries, and 10-ks," *The Journal of Finance*, vol. 66, no. 1, pp. 35–65, 2011.

[18] T. H. Nguyen, K. Shirai, and J. Velcin, "A novel method for predicting stock price movement using the sentiment from social media," *Expert Systems with Applications*, vol. 42, no. 24, pp. 9603–9611, 2015.

[19] R. P. Schumaker and H. Chen, "Textual analysis of stock market prediction using breaking financial news: The azfin text system," *ACM Transactions on Information Systems*, vol. 27, no. 2, pp. 12:1–12:19, 2009.

[20] M. Hagenau, M. Liebmann, and D. Neumann, "Automated news reading: Stock price prediction based on financial news using context-capturing features," *Decision Support Systems*, vol. 55, no. 3, pp. 685–697, 2013.

[21] X. Ding, Y. Zhang, T. Liu, and J. Duan, "Deep learning for event-driven stock prediction," in *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence*, pp. 2327–2333, 2015.

[22] Z. Hu, W. Liu, J. Bian, X. Liu, and T.-Y. Liu, "Listening to chaotic whispers: A deep learning framework for news-oriented stock trend prediction," *Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining*, pp. 261–269, 2018.

[23] S. Mohan, S. Mullapudi, S. Sammeta, P. Vijayvergia, and D. C. Anastasiu, "Stock price prediction using news sentiment analysis," in *IEEE Big Data Computing Service and Applications*, pp. 205–208, 2019.

[24] Y. Xu and S. B. Cohen, "Stock movement prediction from tweets and historical prices," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1970–1979, 2018.

[25] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," *arXiv preprint arXiv:1706.03762*, 2017.

[26] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 4171–4186, 2019.

[27] G. Inoue, B. Alhafni, N. Baimukan, H. Bouamor, and N. Habash, "The interplay of variant, size, and task type in arabic pre-trained language models," in *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pp. 92–104, 2021.

[28] H. Mao, S. Counts, and J. Bollen, "Predicting financial markets: Comparing survey, news, twitter and search engine data," *arXiv preprint arXiv:1112.1051*, 2011.

[29] S. Albahli, A. Irtaza, T. Nazir, A. Mehmood, A. Alkhalifah, and W. Albattah, "A machine learning method for prediction of stock market using real-time twitter data," *Electronics*, vol. 11, no. 20, p. 3414, 2022.

[30] K. AlKhatib, A. Rabab'ah, M. Al-Ayyoub, and Y. Jararweh, "On the use of arabic tweets to predict stock market changes in the arab world," *International Journal of Advanced Computer Science and Applications*, vol. 7, no. 5, 2016.

[31] T.-T. Ho and Y. Huang, "Stock price movement prediction using sentiment analysis and candlestick chart representation," *Sensors*, vol. 21, no. 23, p. 7957, 2021.

[32] M. P. Cristescu, R. A. Nerisanu, D. A. Mara, and S.-V. Oprea, "Using market news sentiment analysis for stock market prediction," *Mathematics*, vol. 10, no. 22, p. 4255, 2022.

[33] Y. Zhao and G. Yang, "Deep learning-based integrated framework for stock price movement prediction," *Applied Soft Computing*, vol. 133, p. 109921, 2023.

[34] A. Mudinas, D. Zhang, and M. Levene, "Market trend prediction using sentiment analysis: Lessons learned and paths forward," *arXiv preprint arXiv:1903.05440*, 2019.

[35] R. Elbadry, S. Ahmad, A. Heakl, D. Bouch, M. Ahsan, M. AlMahri, M. E. khalil, Y. Wang, S. Lahlou, S. Ananiadou, V. Stoyanov, J. Huang, X. Peng, P. Nakov, and Z. Xie, "SAHM: A benchmark for arabic financial and shari'ah-compliant reasoning," *arXiv preprint arXiv:2604.19098*, 2026.

[36] M. H. Albaqawi, E. M. Albalkhi, J. A. Albaiti, and E. Lopedoto, "LLM-based financial sentiment analysis in arabic: Evidence from saudi markets," *arXiv preprint arXiv:2605.19714*, 2026.